# CoachDesk — Project Status Document

**Prepared:** November 2026 · **Version:** 2.0 · **Project:** CoachDesk SaaS Coaching Institute Management System

---

## 1. Our Agenda

Build **CoachDesk** — a production-grade, multi-tenant SaaS platform for Indian coaching institutes, tuition centres, and training academies. Each institute operates independently while sharing one unified infrastructure, letting them manage students, batches, attendance, fees, exams, staff, and parent communication from a single dashboard.

**Core goals:**

- **Multi-tenant isolation** — one platform, many institutes, zero data leakage
- **Enterprise security** — session + JWT auth, RBAC, audit logging, password reset flows
- **Complete SaaS lifecycle** — public signup  trial  subscription  billing
- **Scalable foundation** — small tuition centres today, multi-branch institutes tomorrow
- **Dual frontends** — server-rendered CI4 admin (live) + React SPA (upcoming)

**Technology stack (locked in):**

- **Backend:** PHP 8.1+ / CodeIgniter 4 (HMVC), MySQL 8, no ORM
- **Auth:** Session (admin UI) · JWT + refresh tokens (API) · Password reset via email tokens
- **Frontend:** React + Vite + TypeScript + Tailwind + shadcn/ui *(upcoming)*
- **Email:** SendInBlue (via `NexusSendEmail` helper) · branded inline-CSS templates
- **PDF:** DomPDF (tenant-branded receipts)
- **Table prefix:** `cd_` · **API base:** `/api/v1` · **Admin base:** `/admin`

---

## 2. What We Have Done Till Now

### Database Layer — 100% Complete

| Item | Status | Details |
|---|---|---|
| **Schema** | Done | 44 tables, all `cd_` prefixed, InnoDB, utf8mb4, signed `INT(11)` |
| **Multi-tenancy** | Done | `tenant_id` on every institute-owned row, `branch_id` where applicable |
| **Indexes & FKs** | Done | Tenant-scoped unique constraints, composite indexes, cascade rules |
| **Soft deletes** | Done | `deleted_at` on all business-history tables |
| **Password reset table** | Done | `cd_password_resets` (hashed tokens, expiry, usage tracking) |
| **Session table** | Done | `cd_ci_sessions` migration |
| **Seed data** | Done | 2 tenants, 3 branches, 8 users, 51 permissions, 12 students, 4 batches, fees, payments, exams, attendance, timetable |
| **Login credentials** | Done | All users `Password@123` (bcrypt) |

## Backend Core — 100% Complete

| Item | Status | Details |
|---|---|---|
| **HMVC structure** | Done | `app/Modules/Api/` + `app/Modules/Admin/` |
| **JWT auth** | Done | Login, refresh, rotation, revoke, logout |
| **Session auth** | Done | Admin UI login/logout with bcrypt |
| **Tenant context** | Done | `TenantContext` library + `TenantFilter` |

| | | |
|---|---|---|
| **RBAC engine** | Done | Role + permission + user-override model, `can()` helper |
| **Admin filter** | Done | `AdminAuthFilter` supports `adminauth:permission` syntax |
| **Permission filter** | Done | `PermissionFilter` for API (JWT-aware) |
| **Audit logging** | Done | `cd_activity_logs` on every sensitive action |
| **Helper library** | Done | `current_user` , `tenant_id` , `can` , `money` , `uuid4` , PHP 7.4 polyfills, `NexusSendEmail` |
| **DashboardService** | Done | Aliased-free queries, PHP 7.4 compatible, KPIs + charts data |

## Admin Panel (Server-Rendered CI4) — 100% Complete

| Module | Status | Notes |
|---|---|---|
| **Login / Logout** | Done | Fully styled, error handling, audit logged |
| **Forgot Password** | Done | Email reset link, 30-min expiry, token hashed, invalidates previous |
| **Reset Password** | Done | Verifies token + expiry + usage, updates password, revokes sessions |
| **Register (Public)** | Done | Lead-capture form emails `info@modulelabs.in` (no DB write) |
| **Dashboard** | Done | KPI cards, monthly collection chart, admissions chart, recent payments, upcoming exams |
| **Super Admin Dashboard** | Done | Platform KPIs (tenants, students, MRR), signups chart, expiring subscriptions |
| **Profile / Change Password** | Done | All roles, verifies current password, revokes all tokens, sends confirmation email |

| | | |
|---|---|---|
| **Students** | Done | List, filter, paginate, admit, edit, delete (soft) |
| **Courses** | Done | List, create, edit |
| **Batches** | Done | List, create, enroll students, view students |
| **Teachers** | Done | List, add |
| **Attendance** | Done | Session list, mark daily attendance with bulk actions |
| **Fees** | Done | Fee summary, collect payment with auto-allocation, partial payment handling |
| **Fee Receipt PDF** | Done | Download + inline preview, tenant-branded, allocation breakdown |
| **Exams** | Done | Schedule exams, enter marks (basic) |
| **Reports** | Done | Fee collection, attendance by batch |
| **Settings** | Done | Institute profile, branding, receipt footer |
| **Branches** | Done | Read-only list (CRUD scoped to Super Admin) |
| **Users** | Done | List, create (invite email), reset password (email), force logout (email) |
| **Tenants (Super Admin)** | Done | List, create (auto-provisions owner + branch + subscription), edit, suspend/activate, soft delete |
| **Plans (Super Admin)** | Done | List, create, edit, toggle active, delete if unused |
| **Subscriptions (Super Admin)** | Done | List subscriptions + payments, renew, cancel |

| | | |
|---|---|---|
| **Support Tickets** | Done | List, change status |
| **Audit Logs** | Done | Filterable viewer |
| **Revenue Report** | Done | Monthly breakdown, plan-wise MRR |
| **Platform Settings** | Done | Signup toggle, support email, trial days, maintenance mode |
| **Broadcast** | Done | Send notification to all tenant users |

## Email System — 100% Complete

| Email | Sent When | Recipient | Template |
|---|---|---|---|
| **Welcome (tenant owner)** | New institute created | Owner | Credentials + “change password” CTA |
| **Welcome (new user)** | Owner creates user | New user | Login URL + temp password + role |
| **Password changed** | User self-changes password | User | IP + device + timestamp + “not you?” warning |
| **Admin password reset** | Super Admin resets | User | Temp password highlighted, change required |
| **Force logout notice** | Super Admin revokes sessions | User | Timestamp + “contact admin” |
| **Forgot password link** | User requests reset | User | Reset button + 30-min expiry warning |
| **Reset confirmation** | Password reset completed | User | Timestamp + IP + security advisory |
| **New signup request** | Public form submitted | `info@modulelabs.in` | Full lead detail with reply button |

| | | | |
|---|---|---|---|
| **Signup confirmation** | Public form submitted | Prospect | Thank you + 24-hour response promise |

All templates: **inline CSS**, Gmail/Outlook/Apple Mail safe, HTML-escaped, responsive, gradient header, footer copyright.

## Security & Testing — 100% Complete

| Feature | Status |
|---|---|
| Password hashing (bcrypt) | |
| CSRF protection on all forms | |
| Tenant isolation on every query | |
| Permission check on every route | |
| Force-logout on password change / suspend | |
| Audit trail (logins, payments, tenant changes) | |
| Session regeneration on login | |
| Password reset with hashed tokens + expiry | |
| Email enumeration protection | |
| Cross-tenant access tests | Verified |

## Summary

| Layer | Progress |
|---|---|
| Database | **100%** |
| Backend core | **100%** |
| Admin panel | **100%** |
| Email system | **100%** |
| Security | **100%** |
| **MVP (server-rendered) — Ready for beta use** | |

---

# 3. Future Tasks To Do

## Phase A — React Frontend (Next Priority, ~4–6 weeks)

| Task | Priority | Est. |
|---|---|---|
| Vite + React + TS + Tailwind + shadcn/ui scaffold | Critical | 3 days |
| Auth flow (login, JWT storage, axios interceptor, protected routes) | Critical | 3 days |
| Layout shell (sidebar, topbar, breadcrumbs, dark mode) | Critical | 2 days |
| Dashboard page (KPI cards, charts, widgets) | Critical | 4 days |
| Students CRUD (list, filter, paginate, admit, edit) | Critical | 5 days |
| Courses / Subjects CRUD | High | 3 days |
| Batches CRUD + enrollment | High | 3 days |
| Teachers CRUD | High | 2 days |
| Attendance (mark, view, reports) | High | 4 days |
| Fees (collect, receipt download, defaulters) | High | 5 days |
| Exams (schedule, marks entry, results) | High | 4 days |
| Reports (fee, attendance, performance) | Medium | 3 days |
| Settings (profile, branding, users) | Medium | 2 days |
| React API integration + tenant scoping tests | Critical | 2 days |

## Phase B — Fee Module Deepening (~1–2 weeks)

| Task | Priority |
|---|---|
| Fee structure builder (admission/monthly/quarterly/annual) | High |
| Installment schedule generator | High |
| Bulk payment import (Excel) | Medium |
| Refund workflow with approval | Medium |

| | |
|---|---|
| Fee defaulter list + reminder emails / WhatsApp | High |
| Scholarship / discount / late fee rules | Medium |

### Phase C — Exams & Results (~1–2 weeks)

| Task | Priority |
|---|---|
| Multi-subject exam configuration | High |
| Bulk marks entry (Excel import) | High |
| Auto-grade + rank calculation | High |
| Report card PDF generation | High |
| Merit list + subject-wise analytics | Medium |
| Online MCQ exam (question bank) | Future |

### Phase D — Communication (~1 week)

| Task | Priority |
|---|---|
| SMS integration (MSG91 / Twilio) | High |
| WhatsApp Business API integration | High |
| Absence notifications to parents | High |
| Fee due reminders (cron) | High |
| Notification center UI (bell icon) | Medium |
| Email templates per institute | Medium |

### Phase E — Student & Parent Portals (~2 weeks)

| Task | Priority |
|---|---|
| Student login + portal (attendance, fees, results, timetable) | High |
| Parent login + multi-child dashboard | High |
| Homework submission + study materials download | Medium |

| | |
|---|---|
| Online fee payment (Razorpay / Stripe) | High |
| Mobile-responsive parent experience | High |

## Phase F — SaaS Billing & Launch (~2 weeks)

| Task | Priority |
|---|---|
| Payment gateway integration (Razorpay) | Critical |
| Subscription checkout (plan payment activation) | Critical |
| Auto-renew + grace period | High |
| Usage limits (student/staff/branch caps) | High |
| Invoice generation for subscriptions | High |
| Trial expiry warnings (email + in-app) | High |
| Maintenance mode enforcement | Medium |

## Phase G — DevOps & Production (~1 week)

| Task | Priority |
|---|---|
| Production environment (VPS / Render / Railway) | Critical |
| MySQL managed database + automated backups | Critical |
| Cloud storage for student documents (S3 / R2) | High |
| CI/CD pipeline (GitHub Actions) | High |
| Error monitoring (Sentry) | High |
| **PHP 8.2 upgrade** (required before public launch) | Critical |
| Performance tuning (indexes, caching) | Medium |
| Load testing (1000 concurrent users) | Medium |

## Phase H — Advanced Features (Post-launch)

| Task | Priority |
|---|---|

| | |
|---|---|
| QR-code attendance | Future |
| Biometric integration | Future |
| AI-powered performance analytics | Future |
| Multi-currency support | Future |
| White-label deployments | Future |
| Mobile app (React Native) | Future |
| Library / Inventory / Hostel modules | Future |

---

## Key Metrics & Milestones

| Milestone | Target Date | Status |
|---|---|---|
| **Backend + Admin panel MVP** | **Nov 2026** | **Complete** |
| React frontend v1 | Dec 2026 | Next |
| Beta launch (5 institutes) | Jan 2027 | Planned |
| Public launch (Razorpay billing live) | Feb 2027 | Planned |
| 100 paying institutes | Jun 2027 | Planned |

---

## Dependencies & Risks

| Risk | Mitigation |
|---|---|
| **SendInBlue email quota** | Fallback SMTP provider; monitor daily sends |
| **WhatsApp API rate limits** | Queue system (Redis + worker) |
| **Payment gateway compliance** | PCI-DSS via Razorpay hosted checkout |
| **Multi-tenancy leak** | Automated cross-tenant tests in CI |
| **Data privacy (Indian DPDP Act)** | Consent capture, data export, retention policy |

| | |
|---|---|
| **PHP 7.4 fallback** | Polyfills in place; **upgrade to PHP 8.2 required before public launch** |
| **Scaling DB >2B rows** | Migrate `INT(11)` → `BIGINT` when approaching limit |
| **DomPDF rendering** | Test at production scale; fallback to wkhtmltopdf if needed |

---

## Immediate Next Steps

1. **Start React frontend scaffold** — dashboard + auth first (proves the stack)
2. **Set up cloud storage** (S3 / R2) — student documents need persistence
3. **Add Razorpay sandbox** — get subscription checkout working end-to-end
4. **Write README + deployment guide** for beta testers
5. **Upgrade PHP to 8.2** before beta signups begin
6. **Set up GitHub Actions** — automated tests on every push

---

**Bottom line:** The **complete backend + admin panel is production-ready**. CoachDesk can onboard real institutes **today** using the server-rendered admin panel — including public signup, password reset, tenant provisioning, fee receipts, and platform billing infrastructure. The remaining work is extending to React, portals, and payment gateway automation — all additions to a proven core.

---

*Prepared by: Development Team · For questions: info@modulelabs.in*